

Daily Literature Review — 2026-09-07

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****Topic:**** AI and machine learning in protein function prediction

****Research question:**** How are machine-learning methods being used to predict protein function, what datasets and evaluation metrics are used, and what limitations remain?

Note

AI drafting was unavailable or failed. This is a structured evidence list only. Always read the original papers before using any claims.

Papers

[1] AI-Driven Deep Learning Techniques in Protein Structure Prediction.

- Year: 2024
- Source: PubMed
- Authors: Chen L, Li Q, Nasif KFA, Xie Y, Deng B, Niu S, Pouriyeh S, Dai Z
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/39125995/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[2] Machine learning-based reproducible prediction of type 2 diabetes subtypes.

- Year: 2024
- Source: PubMed
- Authors: Tanabe H, Sato M, Miyake A, Shimajiri Y, Ojima T, Narita A, Saito H, Tanaka K
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/39168869/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[3] Rapid directed evolution guided by protein language models and epistatic interactions.

- Year: 2026
- Source: PubMed

- Authors: Tran VQ, Nemeth M, Bartie LJ, Chandrasekaran SS, Fanton A, Moon HC, Hie BL, Konermann S
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/41712694/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[4] AKT and EZH2 inhibitors kill TNBCs by hijacking mechanisms of involution.

- Year: 2024
- Source: PubMed
- Authors: Schade AE, Perurena N, Yang Y, Rodriguez CL, Krishnan A, Gardner A, Loi P, Xu Y
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/39385030/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[5] Dynamic Prediction of Non-Neutral SARS-Cov-2 Variants Using Incremental Machine Learning.

- Year: 2022
- Source: PubMed
- Authors: Nicora G, Marini S, Salemi M, Bellazzi R
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/35612170/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[6] LCN2 drives ferroptosis-associated ischemia-reperfusion injury after renal transplantation: integrated machine learning and in vivo validation.

- Year: 2025
- Source: PubMed
- Authors: Wu Z, Yu B, He Q, Huang C
- DOI: Not supplied

- URL: <https://pubmed.ncbi.nlm.nih.gov/41205031/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[7] Machine Learning Approaches for Metalloproteins.

- Year: 2022
- Source: PubMed
- Authors: Yu Y, Wang R, Teo RD
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/35209064/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[8] Discovery of natural ROR γ t inhibitor using machine learning, virtual screening, and in vivo validation.

- Year: 2026
- Source: PubMed
- Authors: Yoo H, Han SJ, Lee JE, Cho C, Hong D, Jeong B, Kim S, Jung GY
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/40915561/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[9] Deciphering the mechanism of baicalein in cervical cancer via bioinformatics, machine learning and computational simulations: PIM1 and CDK2 are key target proteins.

- Year: 2025
- Source: PubMed
- Authors: Wang S, Liu C, Ye D, Qi J, Xing Y, Wang J, Fan X, Li X
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/40339869/>
- Key finding: ****[READ AND VERIFY]****

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[10] Machine Learning Prediction of 90-Day Mortality in HBV-Related ACLF Using Olink-Derived Inflammatory Protein Signatures.

- Year: 2025

- Source: PubMed

- Authors: Zhang Y, Sun L, Liu H, Shi K, Lu Y, Feng Y, Wang X

- DOI: Not supplied

- URL: <https://pubmed.ncbi.nlm.nih.gov/41255250/>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[11] AI-Based Protein Structure Prediction in Drug Discovery: Impacts and Challenges.

- Year: 2022

- Source: PubMed

- Authors: Schauperl M, Denny RA

- DOI: Not supplied

- URL: <https://pubmed.ncbi.nlm.nih.gov/35727311/>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[12] Combining Molecular Dynamics and Machine Learning to Predict Drug Resistance Causing Variants of BRAF in Colorectal Cancer.

- Year: 2025

- Source: PubMed

- Authors: Xie L, Lockhart C, Klimov DK, Jafri MS

- DOI: Not supplied

- URL: <https://pubmed.ncbi.nlm.nih.gov/40942081/>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[13] Machine Learning Methods in Protein-Protein Docking.

- Year: 2024
- Source: PubMed
- Authors: Michalik I, Kuder KJ
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/38987466/>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[14] Artificial intelligence guided discovery of a barrier-protective therapy in inflammatory bowel disease.

- Year: 2021
- Source: PubMed
- Authors: Sahoo D, Swanson L, Sayed IM, Katkar GD, Ibeawuchi SR, Mittal Y, Pranadinata RF, Tindle C
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/34253728/>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[15] Machine learning driven prediction of drug efficacy in lung cancer: based on protein biomarkers and clinical features.

- Year: 2025
- Source: PubMed
- Authors: Li J, Chen A, Liu Z, Wei S, Zhang J, Chen J, Shi C
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/40355026/>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[16] Hierarchical graph learning for protein-protein interaction.

- Year: 2023

- Source: PubMed
- Authors: Gao Z, Jiang C, Zhang J, Jiang X, Li L, Zhao P, Yang H, Huang Y
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/36841846/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[17] Sequence-based virtual screening using transformers.

- Year: 2025
- Source: PubMed
- Authors: Zhang S, Huo D, Horne RI, Qi Y, Pujalte Ojeda S, Yan A, Vendruscolo M
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/40721411/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[18] Virtual patient analysis identifies strategies to improve the performance of predictive biomarkers for PD-1 blockade.

- Year: 2024
- Source: PubMed
- Authors: Arulraj T, Wang H, Deshpande A, Varadhan R, Emens LA, Jaffee EM, Fertig EJ, Santa-Maria CA
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/39467131/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[19] Computational modelling of olfactory receptors.

- Year: 2025
- Source: PubMed
- Authors: Odoemelam CS, Steuber V, Schmuker M

- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/40441539/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[20] Novel machine learning approaches revolutionize protein knowledge.

- Year: 2023
- Source: PubMed
- Authors: Bordin N, Dallago C, Heinzinger M, Kim S, Littmann M, Rauer C, Steinegger M, Rost B
- DOI: Not supplied
- URL: <https://pubmed.ncbi.nlm.nih.gov/36504138/>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[21] Highly accurate protein structure prediction with AlphaFold

- Year: 2021
- Source: OpenAlex
- Authors: John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates
- DOI: 10.1038/s41586-021-03819-2
- URL: <https://doi.org/10.1038/s41586-021-03819-2>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[22] Accurate structure prediction of biomolecular interactions with AlphaFold 3

- Year: 2024
- Source: OpenAlex
- Authors: Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore
- DOI: 10.1038/s41586-024-07487-w

- URL: <https://doi.org/10.1038/s41586-024-07487-w>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[23] A guide to machine learning for biologists

- Year: 2021
- Source: OpenAlex
- Authors: Joe G. Greener, Shaun M. Kandathil, Lewis Moffat, David T. Jones
- DOI: 10.1038/s41580-021-00407-0
- URL: <https://doi.org/10.1038/s41580-021-00407-0>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[24] Review of deep learning: concepts, CNN architectures, challenges, applications, future directions

- Year: 2021
- Source: OpenAlex
- Authors: Laith Alzubaidi, Jinglan Zhang, Amjad J. Humaidi, Ayad Q. Al-Dujaili, Ye Duan, Omran Al-Shamma, José Santamaría, Mohammed A. Fadhel
- DOI: 10.1186/s40537-021-00444-8
- URL: <https://doi.org/10.1186/s40537-021-00444-8>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[25] Robust deep learning–based protein sequence design using ProteinMPNN

- Year: 2022
- Source: OpenAlex
- Authors: Justas Dauparas, Ivan Anishchenko, Nathaniel R. Bennett, Hua Bai, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Alexis Courbet
- DOI: 10.1126/science.add2187
- URL: <https://doi.org/10.1126/science.add2187>
- Key finding: ****[READ AND VERIFY]****

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[26] Structure-based protein function prediction using graph convolutional networks

- Year: 2021

- Source: OpenAlex

- Authors: Vladimir Gligorijević, P. Douglas Renfrew, Tomasz Kościółek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor

- DOI: 10.1038/s41467-021-23303-9

- URL: <https://doi.org/10.1038/s41467-021-23303-9>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[27] Swarm Learning for decentralized and confidential clinical machine learning

- Year: 2021

- Source: OpenAlex

- Authors: Stefanie Warnat-Herresthal, Hartmut Schultze, Krishnaprasad Lingadahalli Shastry, Sathyanarayanan Manamohan, Saikat Mukherjee, Vishesh Garg, Ravi Sarveswara, Kristian Händler

- DOI: 10.1038/s41586-021-03583-3

- URL: <https://doi.org/10.1038/s41586-021-03583-3>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[28] Practical guide to SHAP analysis: Explaining supervised machine learning model predictions in drug development

- Year: 2024

- Source: OpenAlex

- Authors: Ana Victoria Ponce Bobadilla, Vanessa Schmitt, Corinna S. Maier, Sven Mensing, Sven Stodtmann

- DOI: 10.1111/cts.70056

- URL: <https://doi.org/10.1111/cts.70056>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[29] AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences

- Year: 2023

- Source: OpenAlex

- Authors: Mihály Váradi, Damian Bertoni, Paulyna Magaña, Urmila Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Sreenath Nair

- DOI: 10.1093/nar/gkad1011

- URL: <https://doi.org/10.1093/nar/gkad1011>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[30] Leakage and the reproducibility crisis in machine-learning-based science

- Year: 2023

- Source: OpenAlex

- Authors: Sayash Kapoor, Arvind Narayanan

- DOI: 10.1016/j.patter.2023.100804

- URL: <https://doi.org/10.1016/j.patter.2023.100804>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[31] Using machine learning approaches for multi-omics data analysis: A review

- Year: 2021

- Source: OpenAlex

- Authors: Parminder Singh Reel, Smarti Reel, Ewan R. Pearson, Emanuele Trucco, Emily Jefferson

- DOI: 10.1016/j.biotechadv.2021.107739

- URL: <https://doi.org/10.1016/j.biotechadv.2021.107739>

- Key finding: **[READ AND VERIFY]**

- Methods/dataset: **[EXTRACT]**

- Limitation: **[EXTRACT]**

[32] Machine Learning in Agriculture: A Comprehensive Updated Review

- Year: 2021
- Source: OpenAlex
- Authors: Lefteris Benos, Aristotelis C. Tagarakis, Georgios Dolias, Remigio Berruto, Dimitrios Katerinis, Dionysis Bochtis
- DOI: 10.3390/s21113758
- URL: <https://doi.org/10.3390/s21113758>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[33] Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction

- Year: 2025
- Source: OpenAlex
- Authors: Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi
- DOI: 10.1101/2025.06.14.659707
- URL: <https://doi.org/10.1101/2025.06.14.659707>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[34] Generative AI

- Year: 2023
- Source: OpenAlex
- Authors: Stefan Feuerriegel, Jochen Hartmann, Christian Janiesch, Patrick Zschech
- DOI: 10.1007/s12599-023-00834-7
- URL: <https://doi.org/10.1007/s12599-023-00834-7>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[35] Small data machine learning in materials science

- Year: 2023

- Source: OpenAlex
- Authors: Pengcheng Xu, Xiaobo Ji, Minjie Li, Wencong Lu
- DOI: 10.1038/s41524-023-01000-z
- URL: <https://doi.org/10.1038/s41524-023-01000-z>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[36] Informed Machine Learning - A Taxonomy and Survey of Integrating Prior Knowledge into Learning Systems

- Year: 2021
- Source: OpenAlex
- Authors: Laura von Rueden, Sebastian Mayer, Katharina Beckh, Bogdan Georgiev, Sven Giesselbach, Raoul Heese, Birgit Kirsch, Michał Walczak
- DOI: 10.1109/tkde.2021.3079836
- URL: <https://doi.org/10.1109/tkde.2021.3079836>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[37] The Role of AI, Machine Learning, and Big Data in Digital Twinning: A Systematic Literature Review, Challenges, and Opportunities

- Year: 2021
- Source: OpenAlex
- Authors: M. Mazhar Rathore, Syed Attique Shah, Dharendra Shukla, Elmahdi Bentafat, Spiridon Bakiras
- DOI: 10.1109/access.2021.3060863
- URL: <https://doi.org/10.1109/access.2021.3060863>
- Key finding: ****[READ AND VERIFY]****
- Methods/dataset: ****[EXTRACT]****
- Limitation: ****[EXTRACT]****

[38] Ensemble deep learning: A review

- Year: 2022
- Source: OpenAlex

- Authors: M. A. Ganaie, Minghui Hu, A. K. Malik, M. Tanveer, Ponnuthurai Nagaratnam Suganthan
- DOI: 10.1016/j.engappai.2022.105151
- URL: <https://doi.org/10.1016/j.engappai.2022.105151>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[39] A review on extreme learning machine

- Year: 2021
- Source: OpenAlex
- Authors: Jian Wang, Siyuan Lu, Shuihua Wang, Yudong Zhang
- DOI: 10.1007/s11042-021-11007-7
- URL: <https://doi.org/10.1007/s11042-021-11007-7>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

[40] Prediction of Chronic Kidney Disease - A Machine Learning Perspective

- Year: 2021
- Source: OpenAlex
- Authors: Pankaj Chittora, Sandeep Chaurasia, Prashant Chakrabarti, Gaurav Kumawat, Tulika Chakrabarti, Zbigniew Leonowicz, Michał Jasiński, Łukasz Jasiński
- DOI: 10.1109/access.2021.3053763
- URL: <https://doi.org/10.1109/access.2021.3053763>
- Key finding: **[READ AND VERIFY]**
- Methods/dataset: **[EXTRACT]**
- Limitation: **[EXTRACT]**

Researcher checklist

- ☐ Read the original papers
- ☐ Verify every claim and reference
- ☐ Record methods, datasets and limitations
- ☐ Add only evidence-supported statements to the manuscript

